Phase Space Reconstruction-Based Online Roughness Monitoring in Robotic Milling of Aircraft Skin Edge

Han-yu Zhang, Wei Xu, *Member, IEEE*, Qing-yu Peng, Dong-fang Wang, Cheng Jiang, and Wen-long Li, *Member, IEEE*

***Abstract*—In robotic milling, monitoring surface roughness in real-time is essential for maintaining the quality of aircraft skin edges. However, existing techniques often struggle with accuracy and robustness due to the complex conditions during milling. This article introduces a novel approach to real-time predict surface roughness in robotic milling of aircraft skin edges. The method begins by dynamically adjusting vibration signals using the frequency response function to maintain prediction accuracy. Subsequently, the phase space reconstruction is employed to map the vibration signals from time domain into phase space diffeomorphic to the robotic system, thereby uncovering the system's dynamic characteristics. Then, the Gramian angular summation field algorithm is used for feature extraction from the phase space trajectories, followed by classification using a convolutional neural network. A vibration-roughness dataset is constructed based on practical experiments in order to train the proposed model. The predictability of the machining system is demonstrated quantitatively. Furthermore, the model's practicality is validated through online experiments. The prediction results based on the test set confirm that this approach offers significant improvements in prediction accuracy and robustness over existing methods.**

***Index Terms*—Aircraft skin parts, online roughness monitoring, phase space reconstruction, robotic milling.**

Nomenclature

$\mathbf{q}(t)$	Displacement response matrix.
$\mathbf{F}(t)$	Excitation matrix.
$\mathbf{M}$	Mass matrix of the system.
$\mathbf{C}$	Damping matrix of the system.
$\mathbf{K}$	Stiffness matrix of the system.
$\mathbf{q}(\omega)$	Fourier transform expression of $\mathbf{q}(t)$.
$\mathbf{F}(\omega)$	Fourier transform expression of $\mathbf{F}(t)$.
$\mathbf{H}(\omega) = [H_{ij}(\omega)]$	Displacement FRF of the system.
$\mathbf{H}_a(\omega)$	Acceleration FRF of the system.
$k = 1, 2, \ldots, K$	Discrete points along the feed direction.
$H_{ij}^R(\omega)$	Real part of $H_{ij}(\omega)$.
$H_{ij}^I(\omega)$	Imaginary part of $H_{ij}(\omega)$.
ω_m	Modal frequency of the system.
ξ_m	Modal damping of the system.
ε	Variation rate of modal parameters.
Δ_k	Distance of adjacent discrete points.
x_t	Coordinates along feed directions.
L	Entire length of milling area of skin parts.
$\mathbf{a}(\omega)$	Fourier transform expression of $\ddot{\mathbf{q}}(t)$.
$\mathbf{H}_a(\omega)^+$	Moore–Penrose pseudoinverse of $\mathbf{H}_a(\omega)$.
$\mathcal{F}^{-1}\{\cdot\}$	Inverse Fourier transform.
$\mathbf{s}_i$	Phase space vector.
$\mathbf{s}_{\eta(i)}$	Nearest neighbors of $\mathbf{s}_i$.
n	Number of vibration sampling points.
m	Embedding dimension.
τ	Delay time.
$\mathbf{S}$	Phase space trajectory.
$I(P, Q)$	Mutual information.
τ_s	Optimal delay time.
m_s	Optimal embedding dimension.
$R_m(i)$	Euclidean distance between $\mathbf{s}_i$ and $\mathbf{s}_{\eta(i)}$.
$a(i, m)$	Distance ratio between $\mathbf{s}_i$and $\mathbf{s}_{\eta(i)}$.
$E(m)$	Distance ratio within entire phase space $\mathbf{S}$
$E_1(m)$	Variation ratio of $E(m)$.
δ	Upper limit of the derivative of $E_1(m)$.
$\widetilde{s}_i$	Angle representation of $\mathbf{s}_i$.
$\mathbf{s}_0$	Reference vector in phase space.
$\widetilde{S}$	Angle representation of $\mathbf{S}$.

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I. Introduction

ROBOTS are increasingly employed in machining large-scale aircraft skin parts due to their versatility and cost efficiency [1], [2], [3]. A notable example is the robotic milling system developed by Fraunhofer IFAM to manufacture the tail-fin of the Airbus A320 [4], as illustrated in Fig. 1. However,

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Fig. 1. Fraunhofer IFAM develops a robotic milling system for manufacturing the tail-fin of an Airbus 320 aircraft [4].

vibration issues, stemming from the low stiffness and stability of both the robot arm and the flexible skin parts [5], negatively impact robotic milling quality of aircraft skin parts. The surface quality of skin edges is essential for maintaining the aerodynamics, stealth properties, and structural integrity of the aircraft. An excessively rough surface can compromise the tight contact between skin components, impair surface sealing effectiveness, and degrade the aircraft's overall performance. Therefore, predicting surface roughness, especially in real-time during the robotic milling process, is crucial for ensuring high-quality and high efficiency aircraft manufacturing.

Surface roughness is typically measured offline by manually selecting specific points on the machined surface for measurement [6]. This approach, however, is not only time-consuming but also prone to errors and inconsistencies, reducing its overall efficiency. While statistical [7] and mathematical [8], [9] models are often employed to predict roughness offline, these methods generally fail to capture the dynamic changes that occur during the machining process [10]. Consequently, their efficiency, accuracy, and reliability are limited. In contrast, online roughness monitoring enables real-time detection of machining irregularities during robotic milling, leading to enhanced production yield and overall process efficiency.

A. Related Works

Online roughness monitoring is generally achieved indirectly through vibration measurement which can be categorized into noncontact and contact methods. Noncontact methods primarily include acoustic emission [11], [12], laser Doppler vibrometry [13], [14], and three-dimensional (3-D) point tracking with structured light systems [15], [16]. Acoustic emission suffers from low signal-to-noise ratios due to interference from industrial environmental noise. Laser Doppler vibrometry exhibits limited measurement accuracy for low-frequency noise [16] and can only capture vibrations along the normal direction. While 3-D point tracking and structured light systems offer a broader measurement range, they demonstrate lower precision and reduced signal-to-noise ratios [16]. Contact methods, such as accelerometer and dynamometer, cannot directly measure the machining area's vibration but indirectly measures it at other positions in the machining system, such as the fixture [17], spindle [18], and workpiece [19]. During the feeding of milling tool, sensors installed at fixed points cannot discern whether changes in vibrations are caused by alterations in the machining position or by variations in the surface roughness. Liu et al. [20] correlated the cutting position with the monitoring signals of cutting force and acceleration. Yao et al. [21], [22] established a position-dependent amplitude preprocessing function to adjust the amplitude of vibration signals. However, due to the nonlinear transmission characteristics of vibrations, the vibration propagation not only affects the amplitude but also induces phase shifts in vibration signals that remains uncompensated by the aforementioned methodologies.

Signal-based prediction models extract features from the acquired vibration signals through time-domain [26], [27], and time-frequency domain [23], [24], [25] analyses. It mainly includes short-time Fourier transform [23], wavelet transform [24], Hilbert Huang transform [25], Gramian angular field [26], and phase space reconstruction [27]. Recent years, AI-based methods have gained attention for their flexibility, practicality, and stronger feature extraction ability [28]. For instance, Zhao et al. [29] developed an long short-term memory (LSTM) network for estimating material removal rates from vibration acceleration signals. Cao et al. [30] created a sparse Bayesian regression model to monitor machining errors by analyzing real-time milling force, cutting parameters, and positions. Lu et al. [31] used a convolutional neural network (CNN) classifier to link vibration signals with machined surface images, while Li et al. [32] proposed a deep learning model incorporating machining parameters and milling force signals to enhance prediction accuracy. He et al. [33] applied an symmetrized dot pattern (SDP)-CNN framework to accurately predict grinding quality from vibration signals. Existing methods rely on single-channel signal or simple combinations of multichannel signals, employing feature extractors to achieve prediction. These approaches exhibit high sensitivity to noise and lack a comprehensive analysis of the nonlinear characteristics embedded in signals.

B. Contribution

This article presents a comprehensive framework for roughness prediction in robot milling of aircraft skin edges, comprising an FRF-based vibration signal correction method and a phase space reconstruction Gramian angular summation field CNN (PSR-GASF-CNN) roughness monitoring model. The vibration signal correction method compensates for vibration signals acquired by accelerometer mounted at fixed point using frequency response functions (FRFs) of discretized milling segments. Time-domain vibration signals are mapped as trajectories in the phase space through PSR method, with GASF employed to extract features from these trajectories. A CNN classifier then performs recognition and classification, enabling precise prediction of surface roughness. The main contributions of this method are summarized as follows.

1) *Enhanced Accuracy and Robustness:* Introduce vibration signal correction method into roughness prediction to prevent accuracy degradation caused by tool path displacement. Moreover, the GASF method encodes global

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relationships between any two elements in time series. This high-dimensional embedding enhances accuracy and robustness.

2) *Reconstructed Dynamic Characteristics:* The phase space is diffeomorphic to the robotic milling system. It implies that the dynamic characteristics of the robotic milling system are reconstructed through limited observed vibration signals. This facilitates the CNN to extract high-level features.
3) *Validated Vibration Signal Predictability:* The robotic milling system is a complex, nonlinear system with limited research on the predictability of its vibration signals. This article quantitatively demonstrates the predictability of vibration signals, thereby supporting the proposed prediction model.

The rest of this article is organized as follows. Section II presents the correction method of vibration signals. The PSR-GASF-CNN model is described in Section III. Section IV conducts practical application experiments, constructs the vibration-roughness dataset, illustrates the predictability of vibration signals, and analyzes the effectiveness of monitoring model. Finally, Section V concludes this article.

II. FRF-Based Vibration Signal Correction Method

During robotic milling of skin parts, fixed position sensors cannot adapt to changes in machining points. To overcome this challenge, we propose a vibration signal correction method. By discretizing the skin edge and separately measuring the FRFs between each discretized point and the accelerometer, position-dependent correction can be applied to the vibration signals. This method facilitates precise monitoring of roughness.

A. Vibration Transmission Model

In the robotic milling of large aircraft skin parts, clamping methods such as magnets and vacuum suction cups are often used to accommodate the scale and complex curvatures of these components. However, these methods do not meet the boundary conditions required for accurate theoretical analysis in dynamics [34]. To address the difficulties in creating precise vibration transmission models for roughness monitoring, this section introduces a vibration transmission model based on the vibration FRF. This approach provides an alternative to theoretical methods, enabling more reliable acquisition of vibrations during robotic milling of aircraft skin parts.

Fig. 2 presents a vibration transmission model of skin parts edge milling. For aircraft skin parts, the stiffness in the $x-$ and $y-$ directions are much greater than that in the z-direction. However, the clamping method based on magnets and vacuum suction cups results in lower clamping stiffness. During actual machining processes, significant vibrations exist in all three directions, x, y, and z. Hence, the transmission model needs to consider vibrations in all three directions.

During the robotic milling process of aircraft skin parts' edge, the position of the milling tool varies over time. However, within a sampling period, the movement distance of milling tool is much smaller than the size of the workpiece. As a result, during each sampling period, the excitation point can be considered invariant. Since the position of the acceleration sensor remains fixed, the vibration model can be simplified. This model can be transformed into a three-degree-of-freedom vibration model with a single excitation point and a single response point. The system dynamic equation of the model is calculated as

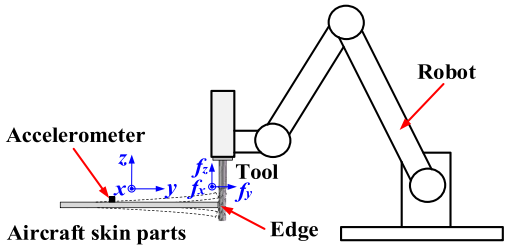


Fig. 2. Vibration transmission model of skin parts edge milling.

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{C}\dot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{F}(t) \tag{1}$$

where $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are, respectively, the mass, damping, and stiffness matrices of the system. $\mathbf{q}(t) = [x(t) \quad y(t) \quad z(t)]^{\mathrm{T}}$ and $\mathbf{F}(t) = [f_x(t) \quad f_y(t) \quad f_z(t)]^{\mathrm{T}}$ represents the displacement response matrix and the excitation matrix in the time domain.

Perform Fourier transform on (1) to obtain

$$\left(-\omega^2\mathbf{M} + \omega j\mathbf{C} + \mathbf{K}\right)\mathbf{q}(\omega) = \mathbf{F}(\omega) \tag{2}$$

where $\mathbf{q}(\omega)$ and $\mathbf{F}(\omega)$ are the Fourier transform expressions of $\mathbf{q}(t)$ and $\mathbf{F}(t)$, respectively.

Define $\mathbf{H}(\omega) = \left(-\omega^2\mathbf{M} + \omega j\mathbf{C} + \mathbf{K}\right)^{-1}$ as the displacement FRF of the system. Then, (2) can be rewritten as

$$\mathbf{q}(\omega) = \mathbf{H}(\omega)\mathbf{F}(\omega). \tag{3}$$

The expanded form of the $\mathbf{H}(\omega)$ matrix is as follows:

$$\mathbf{H}(\omega) = \begin{bmatrix} H_{xx}(\omega) & H_{xy}(\omega) & H_{xz}(\omega) \\ H_{yx}(\omega) & H_{yy}(\omega) & H_{yz}(\omega) \\ H_{zx}(\omega) & H_{zy}(\omega) & H_{zz}(\omega) \end{bmatrix} \tag{4}$$

where the diagonal elements represent the direct FRF of the dynamic system, and the OFF-diagonal elements represent the cross-FRF. Due to the orthogonality of x, y, and z coordinates, the direct FRF are generally much larger than the cross-FRF. Thus, $\mathbf{H}(\omega)$ can be approximated as a diagonal matrix in normal circumstances. However, in actual modal test, we found that the cross-FRF is too obvious to ignore. This means that nonlinearity caused by magnetic chucks clamping methods makes $\mathbf{H}(\omega)$ no longer be a diagonal matrix.

B. Position-Dependent Correction

During the milling process of skin edges, the position-dependent correction method is crucial for reducing the influence of milling position variations on model prediction accuracy. Although the workpiece and the clamping system exhibit inherent nonlinearities, their approximate continuity is maintained. This means, by discretizing the machining area and conducting

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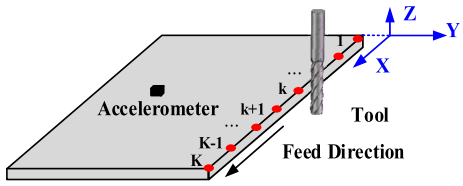


Fig. 3. Discrete spatial model of aircraft skin parts.

modal impact tests on discrete points, the FRF of the entire machining area of the skin parts can be approximated.

Fig. 3 presents a discrete spatial model of skin parts. When a machining tool feeds along x_t-direction, (1) can be rewritten

$$\mathbf{M}_k\ddot{\mathbf{q}}(t)+\mathbf{C}_k\dot{\mathbf{q}}(t)+\mathbf{K}_k\mathbf{q}(t)=\mathbf{F}_k(t) \tag{5}$$

where $k=1,2,\ldots,K$ represents the discrete points along the feed direction. Equation (3) can be rewritten as

$$\mathbf{q}(\omega)=\mathbf{H}_k(\omega)\mathbf{F}_k(\omega). \tag{6}$$

To determine the discrete points, the real part $H_{ij}^R(\omega)$ and imaginary part $H_{ij}^I(\omega)$ of $\mathbf{H}(\omega)$ are used to identify the system modal frequency and modal damping, respectively, as follows:

$$H_{ij}(\omega)=H_{ij}^R(\omega)+iH_{ij}^I(\omega) \tag{7}$$

where $H_{ij}(\omega)$ represents the FRF of excitation at ith degree of freedom (DOF) and response at jth DOF.

The modal frequency ω_m of the system can be determined as

$$\left.\frac{dH^I(\omega)}{d\omega}\right|_{\omega=\omega_m}=0. \tag{8}$$

The modal damping ξ_m of the system can be determined by the following equation:

$$\xi_m=\frac{|\omega_a-\omega_b|}{\omega_m} \tag{9}$$

where ω_a and ω_b satisfy the following equation:

$$\left.\frac{dH^R(\omega)}{d\omega}\right|_{\omega=\omega_a,\omega_b}=0. \tag{10}$$

Therefore, discrete points can be determined by limiting the changes in modal frequency and modal damping

$$\frac{\omega_m^k-\omega_m^{k+1}}{\omega_m^k}\le\varepsilon \quad or \quad \frac{\xi_m^k-\xi_m^{k+1}}{\xi_m^k}\le\varepsilon \tag{11}$$

where ε denotes the variation rate of modal parameters between adjacent discrete points. In practical implementations, it is typically assigned a value of 0.1.

Define the length of adjacent discrete points is Δ_k

$$\Delta_k=x_{k+1}-x_k. \tag{12}$$

$\mathbf{H}(\omega)$ of the skin parts is given by the following equation:

$$\mathbf{H}(\omega)\,|_{x_t}=\begin{cases}\mathbf{H}_1(\omega), x_t\in[0,\Delta_1)\\ \mathbf{H}_k(\omega), x_t\in\left[\sum_1^{k-1}\Delta_i,\sum_1^{k}\Delta_i\right)\\ \mathbf{H}_K(\omega), x_t\in[L-\Delta_{K-1},L]\end{cases} \tag{13}$$

where x_t denotes the coordinates along feed directions, starting from the initial cutting position of aircraft skin parts, as Fig. 2. L denotes the entire length of milling area of skin parts.

Accelerometers are often used in standard modal impact tests, where the measured results are acceleration FRF $\mathbf{H}_a(\omega)$

$$\mathbf{a}(\omega)=-\omega^2\mathbf{q}(\omega)=\mathbf{H}_a(\omega)\mathbf{F}(\omega) \tag{14}$$

where $\mathbf{a}(\omega)$ is the Fourier transform expression of $\ddot{\mathbf{q}}(t)$. Combining (3) and (14), the relationship between acceleration FRF $\mathbf{H}_a(\omega)$ and displacement FRF $\mathbf{H}(\omega)$ is shown as follows:

$$\mathbf{H}_a(\omega)=-\omega^2\mathbf{H}(\omega). \tag{15}$$

Due to the nonlinearity caused by the clamping methods mentioned earlier, (15) can no longer be regarded as a diagonal matrix, and it is difficult to prove its invertibility. Therefore, the Moore–Penrose pseudoinverse is used to obtain the least squares estimation of the excitation signals

$$\mathbf{F}(\omega)=\mathbf{H}_a(\omega)^+\mathbf{a}(\omega) \tag{16}$$

where $\mathbf{H}_a(\omega)^+$ is the Moore–Penrose pseudoinverse of $\mathbf{H}_a(\omega)$.

The time-domain excitation signals at any position during the edge milling process can be obtained as

$$\mathbf{F}(t)=\mathcal{F}^{-1}\{\mathbf{F}(\omega)\} \tag{17}$$

where $\mathcal{F}^{-1}\{\cdot\}$ is the inverse Fourier transform.

So far, the vibration signals acquired at a fixed position have converted into the excitation signals at the actual machining area, completing the correction of the vibration signal.

III. Online Roughness Monitoring Model

Due to the diffeomorphism, phase space can more directly present the nonlinear dynamic characteristics of robotic machining systems compared to time and frequency domains, which is beneficial for extracting features with high identification [35]. Therefore, establishing a roughness monitoring model based on phase space trajectory is crucial to enhance the predictive performance of the model.

A. Phase Space Reconstruction

The robot-tool-workpiece system is a deterministic complex nonlinear dynamic system, where $\{f(1),f(2),\ldots,f(n)\}$ represents a set of observed values. The variable n denotes the number of vibration sampling points. In this article, it is the excitation sequence obtained in Section II-B. The time-domain signal is shown in Fig. 4(a).

Let m be the embedding dimension and τ be the delay time. A m-dimension phase space vector $\mathbf{s}_i$ can be constructed as

$$\mathbf{s}_i=[f(i),f(i+\tau),\cdots,f(i+(m-1)\tau)]^{\mathrm{T}}. \tag{18}$$

The time series can be transformed into the phase space $\mathbf{S}$

$$\mathbf{S}=\left\{\mathbf{s}_1,\mathbf{s}_2,\cdots,\mathbf{s}_{n-(m-1)\tau}\right\}. \tag{19}$$

Taken's theorem proves that by selecting appropriate embedding dimension m and delay time τ, the reconstructed phase space can be diffeomorphic to the original dynamic system.

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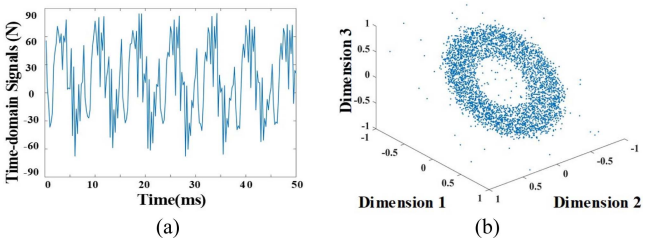


Fig. 4. Vibration signal of robotic milling. (a) Time domain representation. (b) Phase space trajectory.

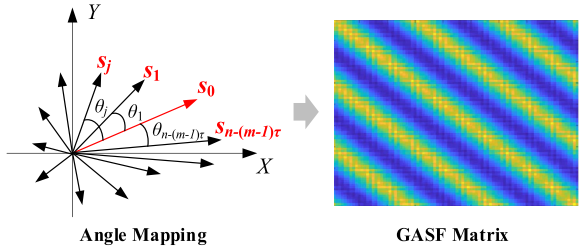


Fig. 5. Basic principle of GASF method.

Since the dynamic system and this phase space are topologically consistent, it is possible to predict the behavior of the robot-tool-workpiece system through phase space.

Define two time series P and Q as

$$P = \{p_1, p_2, \ldots, p_n\} = \{f(1), f(2), \cdots, f(n)\}$$
$$Q = \{q_1, q_2, \ldots, q_n\} = \{f(1+\tau), f(2+\tau), \cdots, f(n+\tau)\} \tag{20}$$

where P represents a time series, and Q is the time series of delay time τ. So, mutual information of P and Q is

$$I(P,Q) = H(Q) + H(P) - H(Q,P) = \sum_i \sum_j P_{pq}(p_i, q_j) \log_2 \frac{P_{pq}(p_i, q_j)}{P_p(p_i) P_q(q_j)} \tag{21}$$

where $H(Q)$ and $H(P)$ represent the information entropy of P and Q, respectively, and $H(Q,P)$ is joint entropy of P and Q. As $I(P,Q)$ increases, so does the correlation strength between P and Q, and conversely. Fraser and Swinney [36] believed that the selection of delay time should minimize correlation. Therefore, we select the optimal delay time τ_s, which minimize

$$\tau_s = \arg\min \{I(P,Q) : 1 \le \tau \le \tau_{\max}\} \tag{22}$$

where $\tau_{\max}$ is the maximum value of τ, to ensure that adjacent points in the phase space are low related.

Assuming $\mathbf{s}_i^m$ is a point in the m-dimensional phase space $\mathbf{S}$, and one of the nearest neighbors of $\mathbf{s}_i^m$ is $\mathbf{s}_{\eta(i)}^m$

$$\eta(i) = \arg\min \left\{ \left\| \mathbf{s}_i^m - \mathbf{s}_j^m \right\| : j = 1, 2, \ldots, n-(m-1)\tau_s, j \ne i \right\}. \tag{23}$$

Using Euclidean distance to measure the distance between two phase points

$$R_m(i) = \left\| \mathbf{s}_i^m - \mathbf{s}_{\eta(i)}^m \right\|, \mathbf{s}_i^m \in \mathbf{S}^m, \mathbf{s}_{\eta(i)}^m \in \mathbf{S}^m. \tag{24}$$

When the dimension of the phase space increases from m to $m+1$, the distance between points $\mathbf{s}_i^{m+1}$ and $\mathbf{s}_{\eta(i)}^{m+1}$ is

$$R_{m+1}(i) = \left\| \mathbf{s}_i^{m+1} - \mathbf{s}_{\eta(i)}^{m+1} \right\|, \mathbf{s}_i^{m+1} \in \mathbf{S}^{m+1}, \mathbf{s}_{\eta(i)}^{m+1} \in \mathbf{S}^{m+1}. \tag{25}$$

Define the distance ratio between $\mathbf{s}_i$and $\mathbf{s}_{\eta(i)}$ in different dimensional spaces as

$$a(i,m) = \frac{R_{m+1}(i)}{R_m(i)}, i = 1, 2, \ldots, n-(m-1)\tau_s. \tag{26}$$

The distance ratio within entire phase space $\mathbf{S}$ is expressed as

$$E(m) = \frac{1}{n - m\tau_s} \sum_{i=1}^{n-m\tau_s} a(i,m). \tag{27}$$

Clearly, $E(m)$ is dependent only on the dimension m. Define

$$E_1(m) = \frac{E(m+1)}{E(m)}. \tag{28}$$

For a deterministic system, when the embedding dimension is greater than a specific value m_s, the phase space structure is fully opened, which can fully reflect the system attractor [37]. That means, $E_1(m)$ will not change after m is greater than the specific value m_s. Generally, we always choose m_s as the optimal system embedding dimension

$$m_s = \min \left\{ m : \frac{dE_1(m)}{dm} = 0 \right\}. \tag{29}$$

In actual processing, there is random noise in the signal, and $E_1(m)$ often is not strictly unchanged. Therefore, assuming a small quantity δ, stop the iteration when $E_1(m)$ is less than δ, and adjust the value of δ according to the actual noise situation

$$m_s = \min \left\{ m : \left| \frac{dE_1(m)}{dm} \right| \le \delta, \delta \ll 1 \right\}. \tag{30}$$

B. PSR-GASF-CNN Roughness Prediction Model

The GASF transformation provides an effective method for converting time series into 2-D feature matrix, enabling feature extraction between any two points in the original time series. Leveraging CNNs' powerful capability in extracting features from 2-D matrix, GASF facilitates the establishment of mapping relationships between vibration signals and surface roughness. The core of GASF lies in converting sequences into angular representations

$$\widetilde{s}_i = \cos\theta_i = \frac{\mathbf{s}_i \cdot \mathbf{s}_0}{|\mathbf{s}_i| \, |\mathbf{s}_0|} \tag{31}$$

where $\mathbf{s}_i$ is an element within the reconstructed phase space trajectory $\mathbf{S}$, $\mathbf{s}_0$ represents a reference vector in phase space, and $\widetilde{s}_i$ is an element within angle series $\widetilde{S} = \{\widetilde{s}_1, \widetilde{s}_2, \cdots, \widetilde{s}_{n-(m-1)\tau}\}$. In general, the reference vector $\mathbf{s}_0$ can be chosen arbitrarily. The angle mapping based on different reference vector $\mathbf{s}_0$ is essentially the same, only providing various forms of feature representation. Fig. 5 demonstrates the image encoding process of vector sequences using GASF.

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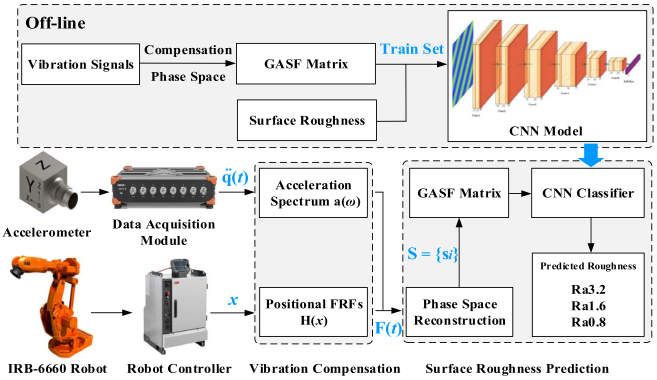


Fig. 6. Architecture of roughness monitoring method.

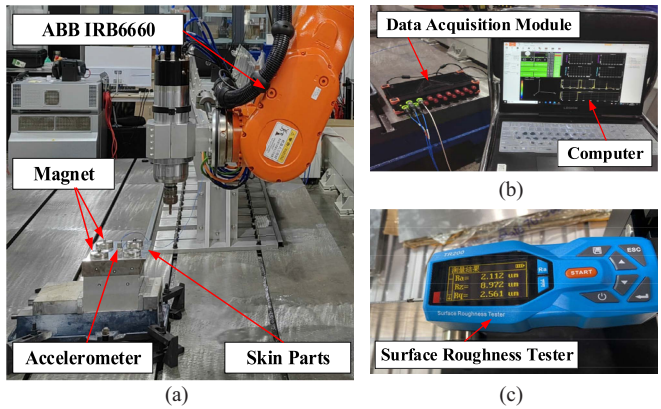


Fig. 7. Practical milling experiment of skin parts. (a) Robotic milling system. (b) Vibration signal acquisition. (c) Roughness measurement.

It is noteworthy that the original GASF transformation was designed for scalar time series, using the ratio of sampled values to the maximum value as input. This approach cannot be directly applied to vector sequences (phase space trajectories). Therefore, we implement a rational generalization by adopting the inner product between phase space vectors and reference vectors as input, thereby extending the applicability of GASF.

After converting the vector sequences into angle series $\widetilde{S}$, the GASF matrix is computed by calculating the cosine of the sum between the angles

$$\text{GASF} = [\cos(\theta_i + \theta_j)] \tag{32}$$

where $i, j = 1, 2, 3, \ldots, n-(m-1)\tau$, respectively.

Given the computationally intensive nature of trigonometric functions, to enhance the calculation efficiency, the above equations are transformed into a matrix operation form

$$\begin{aligned}\text{GASF} &= \left[\widetilde{s}_i \widetilde{s}_j - \sqrt{1-\widetilde{s}_i^2}\sqrt{1-\widetilde{s}_j^2}\right] \\ &= \widetilde{S}^{\mathrm{T}} \cdot \widetilde{S} - \sqrt{I-\left(\widetilde{S}^{\mathrm{T}}\right)^2} \cdot \sqrt{I-\widetilde{S}^2}.\end{aligned} \tag{33}$$

The texture and shape features displayed by the GASF matrix represent the distribution of phase space points and can reflect the nonlinear characteristics of the machining system. Using convolutional neural networks to extract features from the GASF matrix and fitting the nonlinear relationship between the vibration signals and robotic milling surface roughness.

By combining the vibration signal correction and phase space feature extraction, the proposed roughness monitoring method can be described in Fig. 6. It can obtain the position-dependent FRF of the workpiece through offline impulse test. Meanwhile, complete the training of the CNN classifier through a dataset constructed by offline experiments. During online monitoring, the vibration signal $\ddot{\mathbf{q}}(t)$ and the real-time cutting position x_t of the workpiece are obtained through the acceleration sensor and the robot controller, respectively. According to (17), calculate the compensated vibration excitation signal $\mathbf{F}(t)$ and perform phase space reconstruction to obtain the phase space trajectory $\mathbf{S}$. According to (33), calculate the GASF matrix and input it into the CNN classifier to achieve surface roughness online monitoring.

IV. Model Training and Practical Experiment

This section outlines the practical experimental setup designed to construct the vibration-roughness dataset and train the roughness monitoring model. The predictability of vibration signals is quantitatively demonstrated. Compared to similar methods, the proposed method in this article has high accuracy and strong robustness. Online monitoring experiment is conducted to demonstrate the practicality of the prediction model. All the program codes used in the experiments described in the following (unless otherwise specified) are written in MATLAB R2023a and run on a computer with an Intel i7-8750H CPU, a 16GB DDR4 RAM and a NVIDIA GTX1060 GPU.

A. Dataset Construction Based on Practical Experiment

1) Practical Milling Experiment Setup: The surface roughness prediction model will be trained based on a dataset constructed from practical skin parts milling experiment. The selected workpiece for this article is a skin part measuring 200 × 200 × 2 mm, made from 6061 aluminum alloy. Fig. 7 provides an overview of the experimental setup, featuring an ABB IRB-6660 robot for the milling platform. The end effector is an EBLTS TS120F high-speed electric spindle. Acceleration signals, which are monitored using a KSI-138M100 triaxial accelerometer, are acquired by commercial software DewesoftX and SIRIUS X data acquisition module. The workpiece is secured to the vise fixture using strong magnets, with an overhang length of 20 mm. For different experimental groups, the position of the strong magnets is kept constant to minimize errors. The accelerometer is mounted at the geometric center of the skin parts using a strong magnetic base. Edge milling is performed using a three-flute solid carbide flat end mill with a helix angle of 38° and a diameter of 10 mm.

The arithmetic average roughness Ra, which is in accordance with ISO 21920-2:2021, is employed as the quantitative metric for evaluating surface roughness. Surface roughness is measured

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TABLE I
PEARSON CORRELATION ANALYSIS

Parameters	$R_{Pearson}$	P value	Confidence Interval	Cases
Values	0.744	0.003	[0.360, 0.933]	24

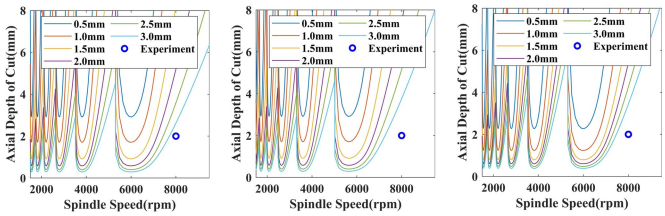


Fig. 8. Stability lobe diagram of robotic machining system at different pose. (a) At the start of the machining trajectory. (b) At the middle of the machining trajectory. (c) At the end of the machining trajectory.

using a contact-type surface roughness tester (TR200). To avoid influences of burrs from machining, measurements are taken along the side midline, perpendicular to thickness direction. The experimental setup for roughness measurement is shown in Fig. 7(c) The roughness measurement methodology follows the international standard ISO 21920-3:2021.

2) Robotic Milling Dataset Construction: Before conducting formal experiments, preliminary research is carried out on the hardware platform shown in Fig. 7. In pre-experiments, multi-parameter combinations are implemented: spindle speed is set at three levels (4000 r/min, 6000 r/min, and 8000 r/min), feed rate is divided into eight gradients from 1 mm/s to 8 mm/s (1 mm/s intervals) and radial depth of cut (RDC) is fixed at 1 mm. The axial depth of cut (ADC) is fixed at 2.0 mm, as dictated by the geometry of the skin parts. To enhance data reliability, 10 discrete measurement points are selected along the edge region of the skin, and arithmetic mean of surface roughness under each parameter combination is calculated.

As shown in Table I, by establishing a Pearson correlation model between feed per tooth and average surface roughness, a significant positive correlation is observed (two-tailed test significance level $p = 0.003$), with the probability of rejecting the null hypothesis below 0.3%. Analysis reveals that increasing spindle speed effectively reduces feed per tooth, thereby improving machining surface quality. Based on this conclusion, spindle speed is fixed at 8000 r/min for the formal experimental phase. As delineated in Table II, feed rate and the RDC are selected as independent variables for constructing the robotic milling dataset. The ADC is fixed at 2.0 mm (dictated by the geometry of skin parts). A full factorial design of experiments is implemented to define feed rate and RDC levels.

To validate the rationality of the dataset design, chatter stability analysis [38] is performed. Three robot postures are selected at the start, middle, and end of the machining trajectory, and their corresponding stability lobe diagrams are shown in Fig. 8(a)–(c). The horizontal axis denotes the spindle speed, while the vertical axis represents the ADC. Stability lobes corresponding to different RDC are distinguished by color-coded curves, with explicit

TABLE II
PROCESS PARAMETERS OF PRACTICAL MILLING EXPERIMENT

Exp No.	Feed Rate (mm/s)	RDC (mm)	Exp No.	Feed Rate (mm/s)	RDC (mm)	Exp No.	Feed Rate (mm/s)	RDC (mm)
1	1	0.5	21	1	1.5	41	1	2.5
2	2	0.5	22	2	1.5	42	2	2.5
3	3	0.5	23	3	1.5	43	3	2.5
4	4	0.5	24	4	1.5	44	4	2.5
5	5	0.5	25	5	1.5	45	5	2.5
6	6	0.5	26	6	1.5	46	6	2.5
7	7	0.5	27	7	1.5	47	7	2.5
8	8	0.5	28	8	1.5	48	8	2.5
9	9	0.5	29	9	1.5	49	9	2.5
10	10	0.5	30	10	1.5	50	10	2.5
11	1	1	31	1	2.0	51	1	3.0
12	2	1	32	2	2.0	52	2	3.0
13	3	1	33	3	2.0	53	3	3.0
14	4	1	34	4	2.0	54	4	3.0
15	5	1	35	5	2.0	55	5	3.0
16	6	1	36	6	2.0	56	6	3.0
17	7	1	37	7	2.0	57	7	3.0
18	8	1	38	8	2.0	58	8	3.0
19	9	1	39	9	2.0	59	9	3.0
20	10	1	40	10	2.0	60	10	3.0

TABLE III
SURFACE ROUGHNESS CLASSIFICATION

Ra (μm)	[0.8, 1.6)	[1.6, 2.0)	[2.0, 2.5)	[2.5, 3.2)	[3.2, 6.3)
	Class 1	Class 2	Class 3	Class 4	Class 5

annotations provided in the legend. Practical experimental parameters are the blue markers, as the spindle speed and ADC are fixed at 8000 r/min and 2.0 mm, respectively. In summary, all machining parameters in the dataset are positioned within stable lobes, which ensures machining stability during the practical experiment.

The parameter ranges cover typical values encountered in aircraft skin milling applications. For each machining parameter, measure surface roughness at 60 discrete points along the edge of the skin workpiece and acquire corresponding vibration signals to construct a vibration-roughness dataset which includes 3600 samples on 3 channels. The surface roughness values are classified into five discrete categories using the R10 preferred number series (per ISO 3:1973) combined with surface roughness gradations defined in ISO 21920, as tabulated in Table III.

B. Training of Surface Roughness Prediction Model

1) Vibration Signal Correction: The modal impulse test via impact hammer excitation enables the determination of the FRF at different positions along the edge of skin parts. Fig. 9 illustrates the setup configuration for the modal impulse test. A KSI-728A002 medium-sized impulse hammer is used in the experiments, with a sensitivity of 2.52 mV/N and an impact force range of 0 to 2 kN. The hammer head has a mass of 100 g and a diameter of 5 mm. Each position is struck three times, and the results are averaged. The FRF data is acquired using

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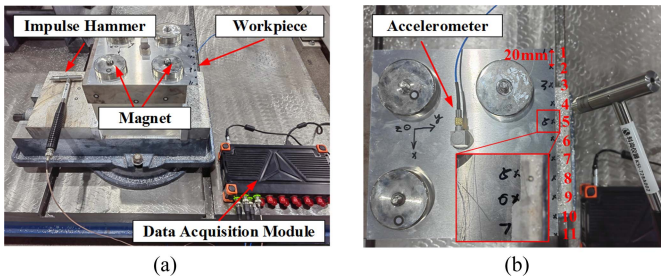


Fig. 9. Modal impulse test setup. (a) Model test experiment. (b) Discretization of the skin edge.

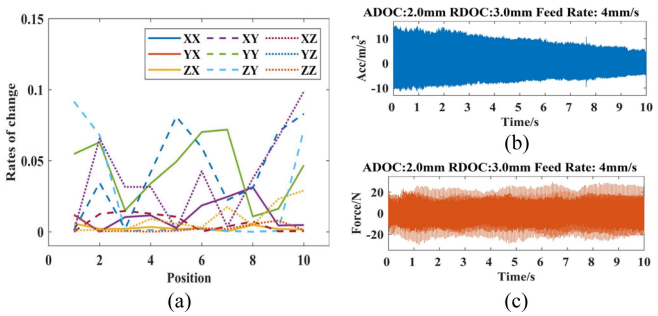


Fig. 10. Vibration signal correction. (a) Variation rates of the first-order modal frequency across adjacent points. (b) Acquired vibration acceleration signal. (c) Compensated vibration force signal.

the commercial software DewesoftX and the SIRIUS X data acquisition module.

During the milling process, due to the cutting action of the milling cutter, the skin part forms a notch that advances along the feed direction. The cutting process of the cutter on the workpiece occurs at this notch, thereby applying milling forces to the workpiece. As shown in Fig. 9(b), a dedicated notch is machined at the skin edge to enable triaxial excitation (X, Y, and Z directions) using an impact hammer at this location. This configuration better simulates practical milling conditions.

The accelerometer is positioned at the geometric center of the skin component, consistent with the machining experiments. A uniform discretization method is employed, dividing the skin edge into 11 equally spaced points along the feed direction, with a spacing interval of 20 mm between adjacent points. Using the measured FRF of the workpiece, adjacent points' first-order modal frequency variation rates in each direction are shown in Fig. 10(a). In the terminal and central regions of skin parts, higher modal frequency variation rates are observed due to increased distance from magnets. Conversely, near the magnets (e.g., around point 3 and 8), enhanced stability corresponds to lower variation rates. The skin exhibits sufficient continuity such that even with the simple uniform discretization method, the modal frequency variation rate between any adjacent points remains below 0.1, satisfying (11) with ε set to 0.1. The discrete FRF approximately represent the full-edge FRF of the skin within the allowable 0.1 tolerance.

To validate the effectiveness of the vibration signal correction model, a vibration signal segment shown in Fig. 10(b) is

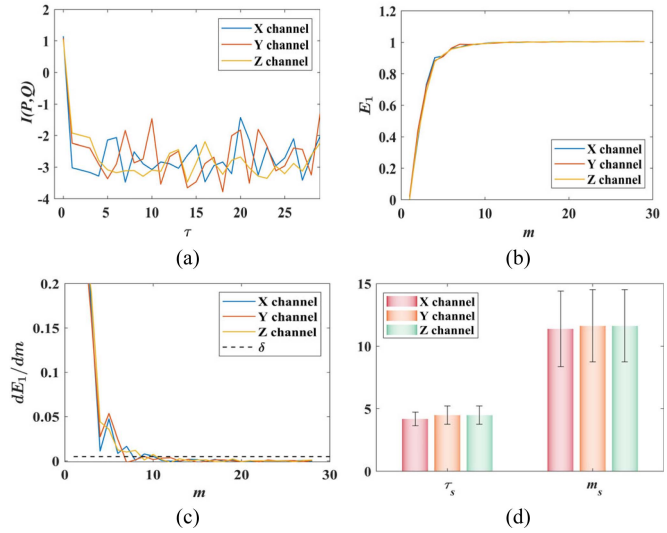


Fig. 11. Selection of critical parameters. (a) Mutual information. (b) Distance ratio. (c) Derivative of distance ratio. (d) Delay time τ_s and embedding dimension m.

selected, where process parameters are annotated in the figure, corresponding to the machining segment between point 8 and point 10. As the cutting tool gradually moves away from the accelerometer, the vibration amplitude significantly decreases. As shown in Fig. 10(c), the vibration amplitudes at different positions become nearly consistent after correction, demonstrating the effectiveness of correction.

2) Phase Space Reconstruction and Predictability Proof: The preset maximum delay time $\tau_{\max}$ and maximum embedding dimension $m_{\max}$ is set to 30 and 50, respectively. According to (21), traverse the range of τ values and calculate the mutual information of X, Y, and Z channel signals. The calculation results are shown in Fig. 11(a). The curve initially decreases and then increases, exhibiting a local minimum around $\tau = 5$.

According to (23)–(28), calculate the distance ratio E_1 for the signals of channels X, Y, and Z, as shown in Fig. 11(b). E_1 increases with the increase of m, but the growth rate gradually slows down and remains unchanged. Taking δ as 0.005, the derivative dE_1/dm is calculated in Fig. 11(c). As can be clearly seen from the figure, when the embedding dimension m is greater than 15, E_1 remains essentially constant, and the dE_1/dm is less than δ, which meets the requirement.

By selecting multiple sets of signals from each combination of processing parameters, we can calculate the optimal delay time τ_s and optimal embedding dimension m_s for the three channels according to (22) and (30), respectively. The mean and standard deviation of these calculations are shown in Fig. 11(d). It can be observed that τ_s and m_s of the data from the three channels are similar. This indicates that the three channels exhibit similar nonlinear characteristics. Therefore, set the optimal delay time τ_s as 5 and the optimal embedding dimension m_s as 15.

Selecting multiple sets of signals from each machining parameter combination, the largest Lyapunov exponent (LLE) and the Kolmogorov entropy (KE) of the X, Y, and Z channel signals are calculated. The mean and standard deviation of LLE and KE,

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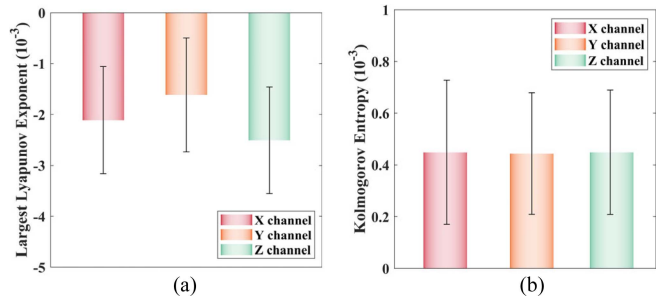


Fig. 12. Parameters for predictability analysis of X, Y, and Z channel signals. (a) Largest Lyapunov exponent. (b) Kolmogorov entropy.

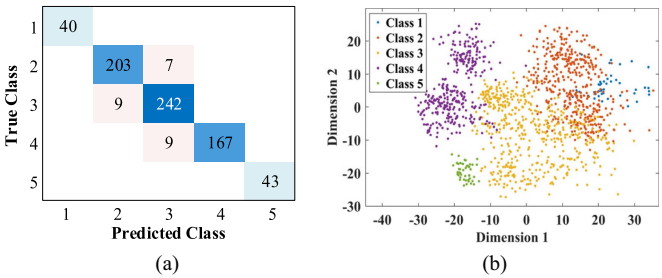


Fig. 13. Training results of roughness prediction model. (a) Confusion matrix chart. (b) Visualization of train set by PCA.

as shown in Fig. 12. LLE represents the average exponential rate at which the phase space trajectory of a nonlinear system converges or diverges over time. The LLE values of signals from all three channels are negative and close to zero, indicating that the robotic milling system is asymptotically stable. This stability implies that the phase space trajectory will ultimately converge to an attractor or fixed point. Meanwhile, the positive KE of the signal indicates the presence of some degree of chaos in the system, but the KE being close to zero suggests that the level of chaos is very low.

In summary, the robot-tool-workpiece system exhibits low chaos, making its dynamic behavior predictable over long periods. The acquired signals can be used to forecast the system's future behavior. The phase space trajectories of the signals form ordered closed loops, and the features of these trajectories can be identified using GASF-CNN, thereby enabling the prediction of surface roughness. This validates the predictability of the proposed roughness prediction model.

3) Training of Roughness Prediction Model: After feature extraction, the final dataset consists of 3600 samples. The training set and test set are divided in an 8:2 ratio, with 2880 samples in the training set and 720 samples in the test set. The prediction model is trained on a computer with an Intel i7-9700 CPU with 3.00 GHz, a 64GB DDR4 RAM, and a NVIDIA RTX3090 GPU with 24 GB VRAM. To improve the prediction model's accuracy, Bayesian optimization is used to fine-tune hyperparameters like the initial learning rate, learning rate drop factor and so on. After 20 iterations, the trained neural network has a classification accuracy of 96.7% for the test set.

TABLE IV
PREDICTION RESULTS OF COMPARISON MODELS

Comparison Models	Test Set Prediction Accuracy			
	SNR +∞	SNR 10	SNR 5	SNR 0
PSR-GASF-CNN (Proposed Model)	96.7%	95.3%	93.2%	90.1%
PSR-CNN	88.2%	85.5%	82.7%	79.5%
GASF-CNN	94.6%	90.4%	87.9%	84.7%
CNN	82.1%	76.9%	65.3%	55.2%
Zhao's	91.1%	88.8%	83.5%	79.3%
Lu's	89.3%	84.2%	78.2%	72.6%
He's	91.2%	87.6%	80.3%	75.6%

The confusion matrix is shown in Fig. 13(a). Specimens are predominantly clustered within the Ra 1.6 μm to Ra 3.2 μm range (accounting for 88% of the dataset), exhibiting limited misclassifications (error rate: 3.3%), indicating that they have similar features. There are fewer samples smaller than Ra 1.6 μm and larger than Ra 3.2 μm, and the classification accuracy of these samples is nearly 100%, indicating that the roughness prediction model performs well in handling extreme categories.

Visualize the classification performance of the training set using the PCA algorithm, as shown in Fig. 13(b). The model performs well in most categories, with samples forming distinct clusters in the 2-D space. Although some overlap exists between different clusters, relatively clear boundaries remain observable. This aligns with practical observations that samples with marginally different roughness values exhibit similar vibration signal characteristics. The limited differentiation between Class 1 and Class 2 samples stems from milling process constraints: most samples below Ra1.6 cluster near Ra1.6, with only a minimal number of samples falling below Ra1.3.

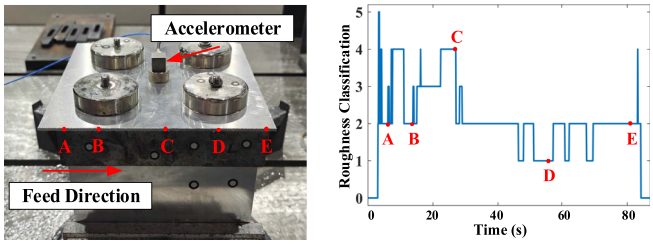
To verify the robustness of the model, Gaussian noise is introduced into the experimental data and SNR is used to represent the strength of the noise

$$\mathrm{SNR} = 10\lg\left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}\right) \tag{34}$$

where P_{signal} and P_{noise} represent the power of signal and noise, respectively. A lower SNR indicates a higher level of noise.

For the proposed PSR-GASF-CNN model in this study, we conducted systematic ablation studies involving three architectural variants: PSR-CNN, GASF-CNN, and CNN baseline. Three SOTA prediction models (Zhao's model [29], Lu's model [31], and He's model [33]) were included for comparative analysis. All seven models were trained using identical methodologies on the dataset constructed in this study, with their prediction performance quantitatively compared in Table IV. The PSR-GASF-CNN model achieves a high prediction accuracy of over 96.7%. This method exhibits strong robustness with minimal susceptibility to noise interference. Even in strong noise environments with a SNR of 0 dB, the model retains 90% classification accuracy. The high precision and strong robustness of the model can be explained by the following two aspects.

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(a) (b)

Fig. 14. Practical online monitoring experiment. (a) Experiment setup. (b) Online roughness monitoring program.

1) Based on Takens' embedding theorem, the phase space and the robot milling system are diffeomorphic. It implies that the dynamic characteristics of the robotic milling system are reconstructed through vibration signals. This facilitates the CNN to extract high-level features, and improves the accuracy and robustness of the predictive model.
2) The GASF method not only captures local variations between adjacent elements but also encodes global temporal relationships between any two elements in the time series. Such high-dimensional embedding enhances classification accuracy and demonstrates strong immunity to random Gaussian noise, which is commonly encountered in engineering applications.

C. Online Monitoring Experiment

To validate the applicability of the proposed roughness monitoring model, a practical online monitoring experiment is conducted using the robotic machining system described in Section IV-A. The use of a variable feed rate machining strategy allows for the generation of different surface roughness levels, which helps analyze the sensitivity of the model to roughness changes. The RDC and ADC are fixed at 2.0 mm. In this strategy, the feed rate for the A-B-C segment is set to 3 mm/s, while for the C-D-E segment, it is set to 1 mm/s, as shown in Fig. 14(a). The edge regions (i.e., before point A and after point E) are excluded from the analysis due to structural instability.

By applying the prediction model trained in the previous section, we are able to perform online monitoring of surface roughness of the skin parts. The predicted results are shown in Fig. 14(b), with points A to E marked. In the predicted results, the B-C segment is located between two magnets, where the clamping stiffness is higher, resulting in lower roughness compared to the A-B segment. At point C, the model accurately responds to the roughness changes caused by the abrupt feed rate change. Point D is closer to the magnet, where the clamping stiffness is optimal, and as a result, this point exhibits the best machining quality.

At each feature point, ten discrete points are selected in the vicinity, and their surface roughness is measured. The results indicate that the model has a higher prediction accuracy in the low-speed segments (nearly 100%) and a lower accuracy in the variable-speed machining segments (only 60%). The overall prediction accuracy is 90%, lower than that of the test set. This discrepancy can be attributed to the randomness and instability inherent in the actual machining process, which affects the model's predictive performance. Overall, the proposed model provides real-time surface roughness monitoring, promptly identifies robotic milling anomalies of aircraft skin edge.

V. Conclusion

We propose a PSR-GASF-CNN-based roughness monitoring model for online surface roughness assessment in aircraft skin edge milling. This model employs position-dependent FRFs to correct vibration signals, effectively suppressing the influence of tool feed motion along the workpiece. The methodology reconstructs vibration signals into phase space through delay embedding. When integrated with CNN classifiers, the global relationships of phase space trajectories are encoded via GASF, achieving over 96.7% accuracy on test sets. The model demonstrates strong robustness, maintaining above 90% accuracy in high-noise environments. Additionally, the predictability of vibration signals is validated through LLE and KE analyses. While current model focuses on rectangular planar workpieces, and future research will extend this framework to free-form surface workpieces. Overall, this solution successfully addresses online roughness monitoring requirements of aircraft skins and offers substantially guidance for industrial implementation.

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